CHROMOSOME CLASSIFICATION USING DEEP LEARNING

AI/ML SUMMER INTERNSHIP PROJECT

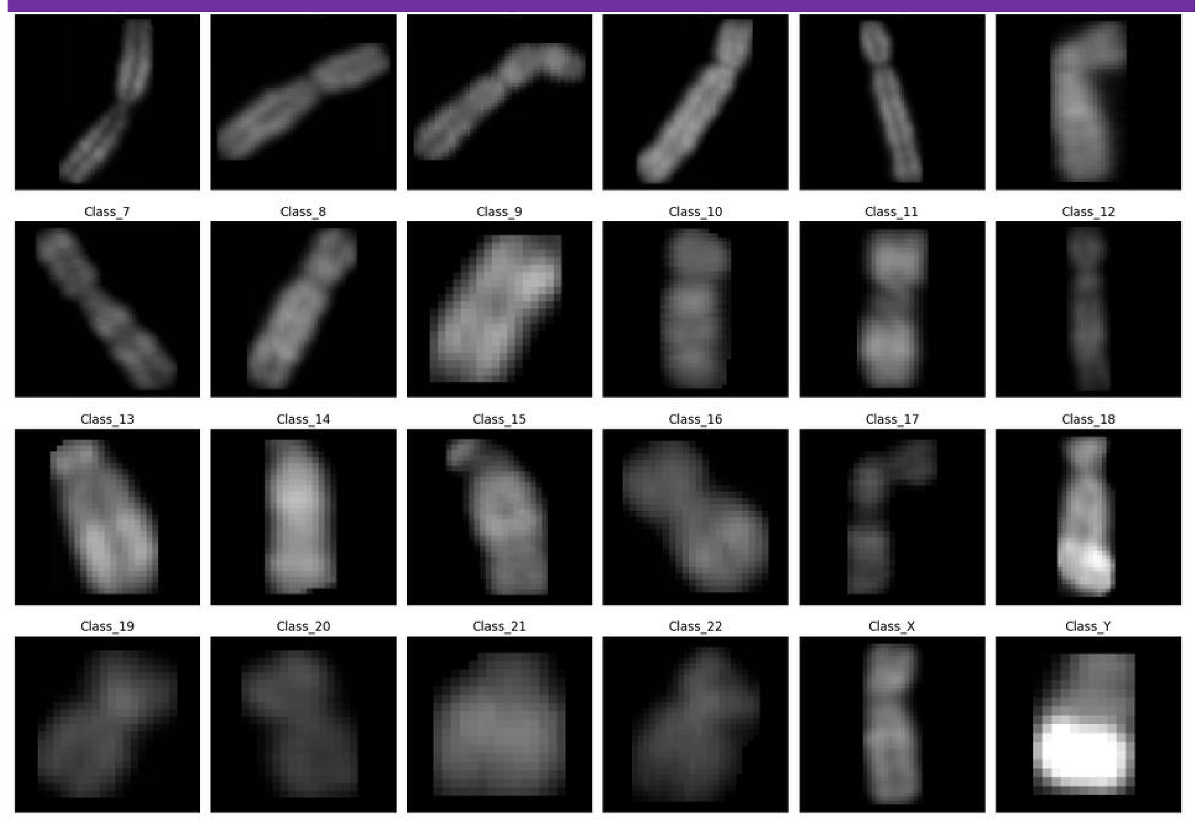
NAME : CHITIKELA RAMBHADRA KUMAR, ROL NO : 202311024 , MENTOR : DR.VENKATA PHANIKRISHNA B

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY VADODARA

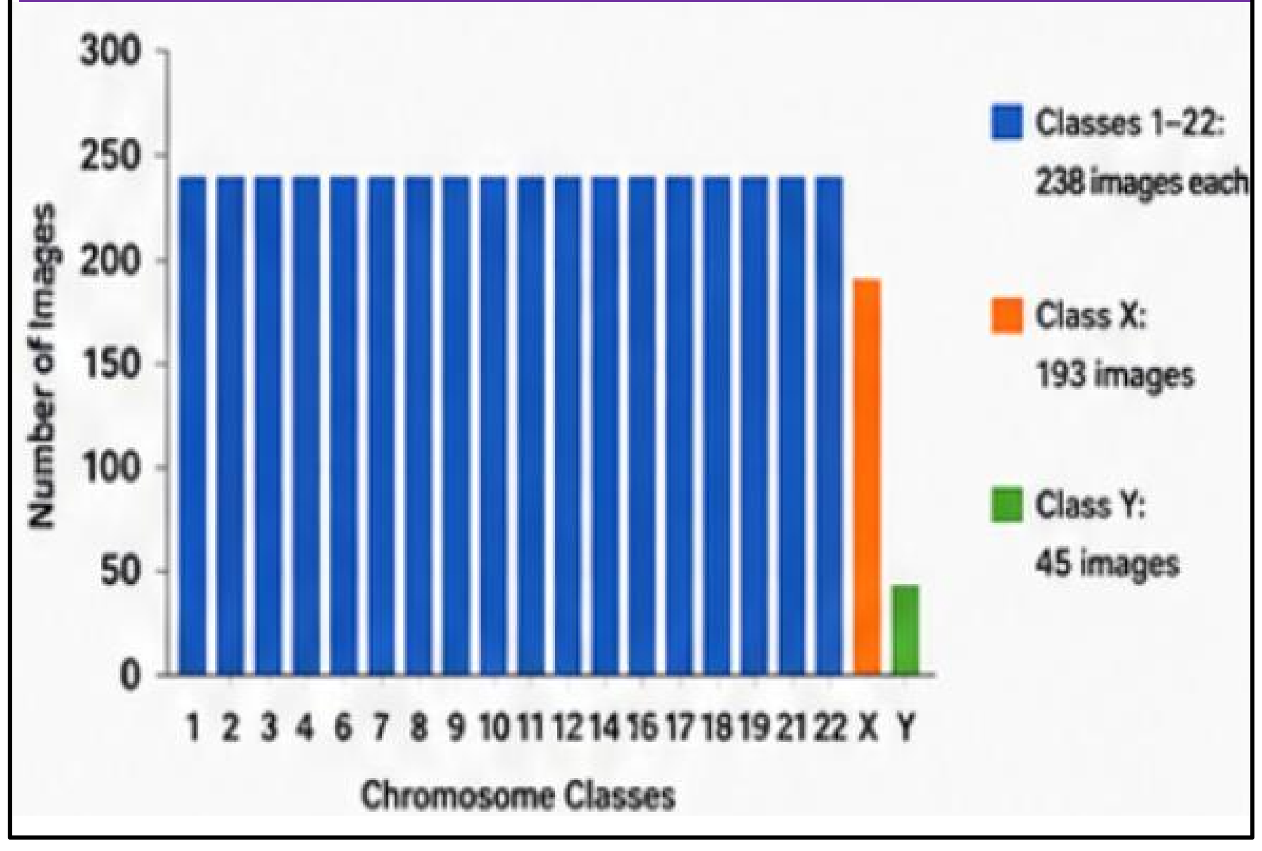
PROJECT OVERVIEW

This project uses deep learning techniques to classify chromosomes into 24 classes (1–22, X, Y). Advanced image preprocessing and state-of-the-art CNN architectures areused to achieve high classification accuracy.

SAMPLE CHROMOSOME IMAGES(24 CLASSES)

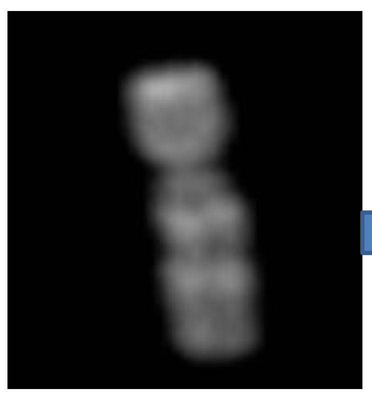


CLASS DISTRIBUTION



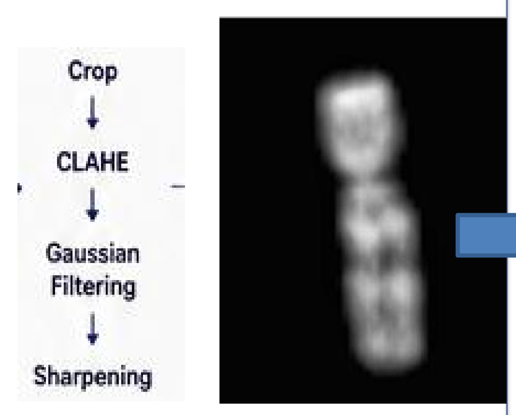
HOW IT WORKS

1 INPUT IMAGE



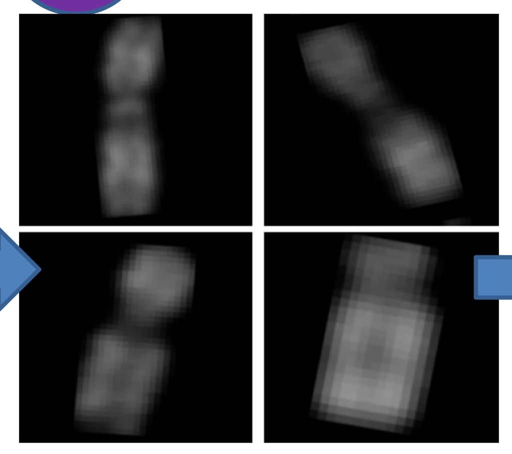
Single Chromosome

2 PREPROCESSING



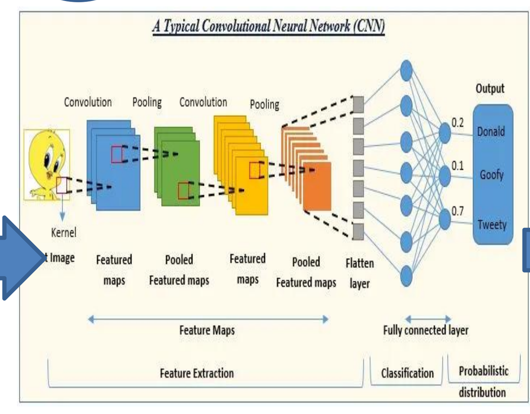
Enhanced inout

3 DATA AUGMENTATION



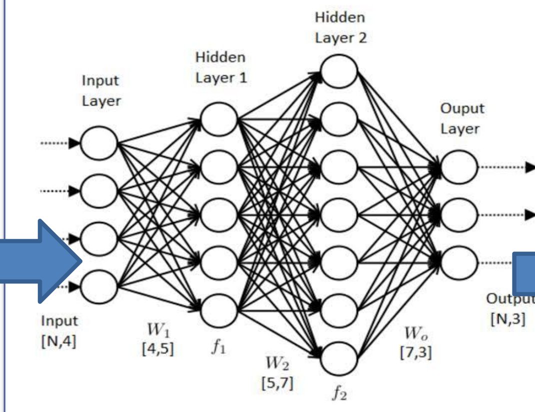
Creating more diverse sample to improve performance

4 FEATURE EXTRACTION



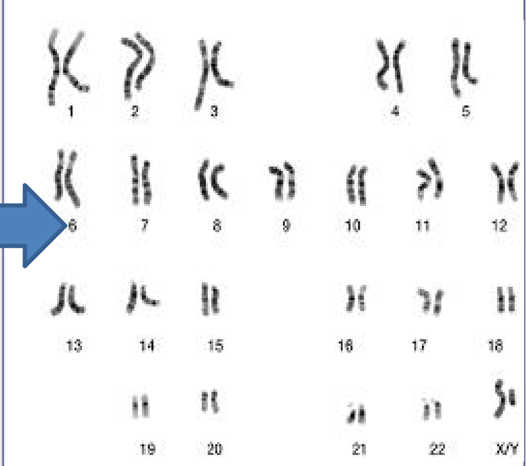
Deep CNN Architectur

5 CLASSIFICATION



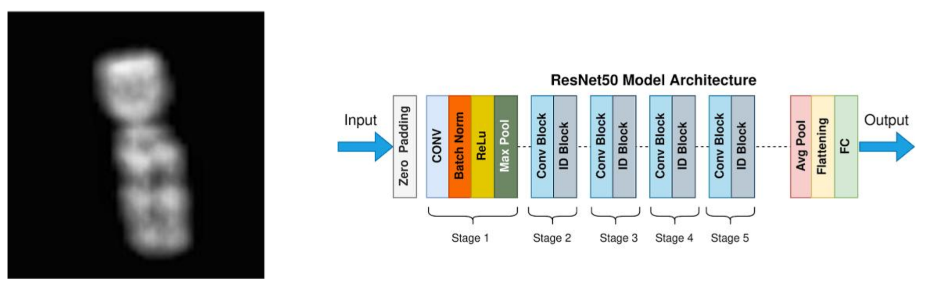
Predicting which class the chromosome belongs to

6 OUTPUT



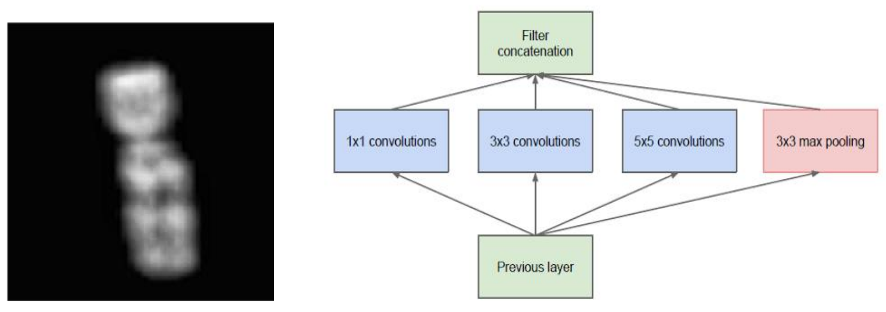
DEEP LEARNING MODELS

RESNET 50



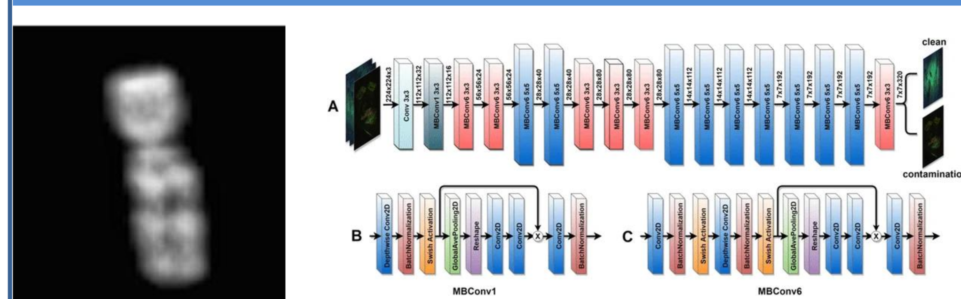
- Skip connections: Uses residual links to bypass layers.
- Deep training: Solves vanishing gradients for deeper networks.
- Standard baseline: Widely used benchmark for image tasks.
- Resource heavy: Large file size and high parameters.

INCEPTIONV3



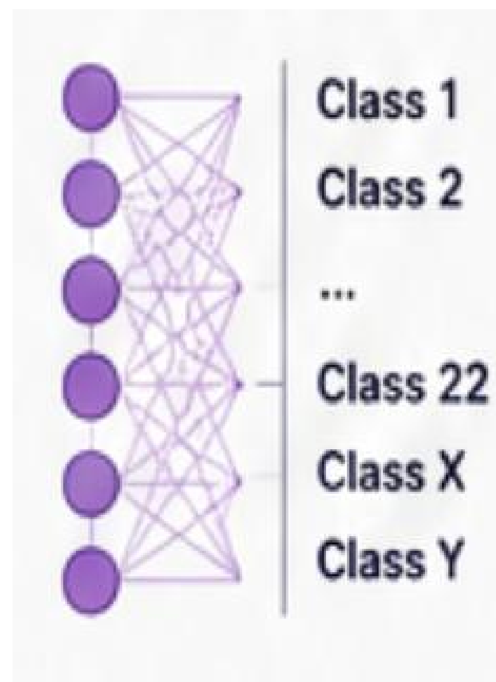
- Multi-scale: Processes different filter sizes simultaneously.
- Factorized filters: Splits large convolutions to save memory.
- High efficiency: Deep architecture with lower computational cost.
- Complex design: Parallel branches make customization difficult.

EFFICIENTNET-B0

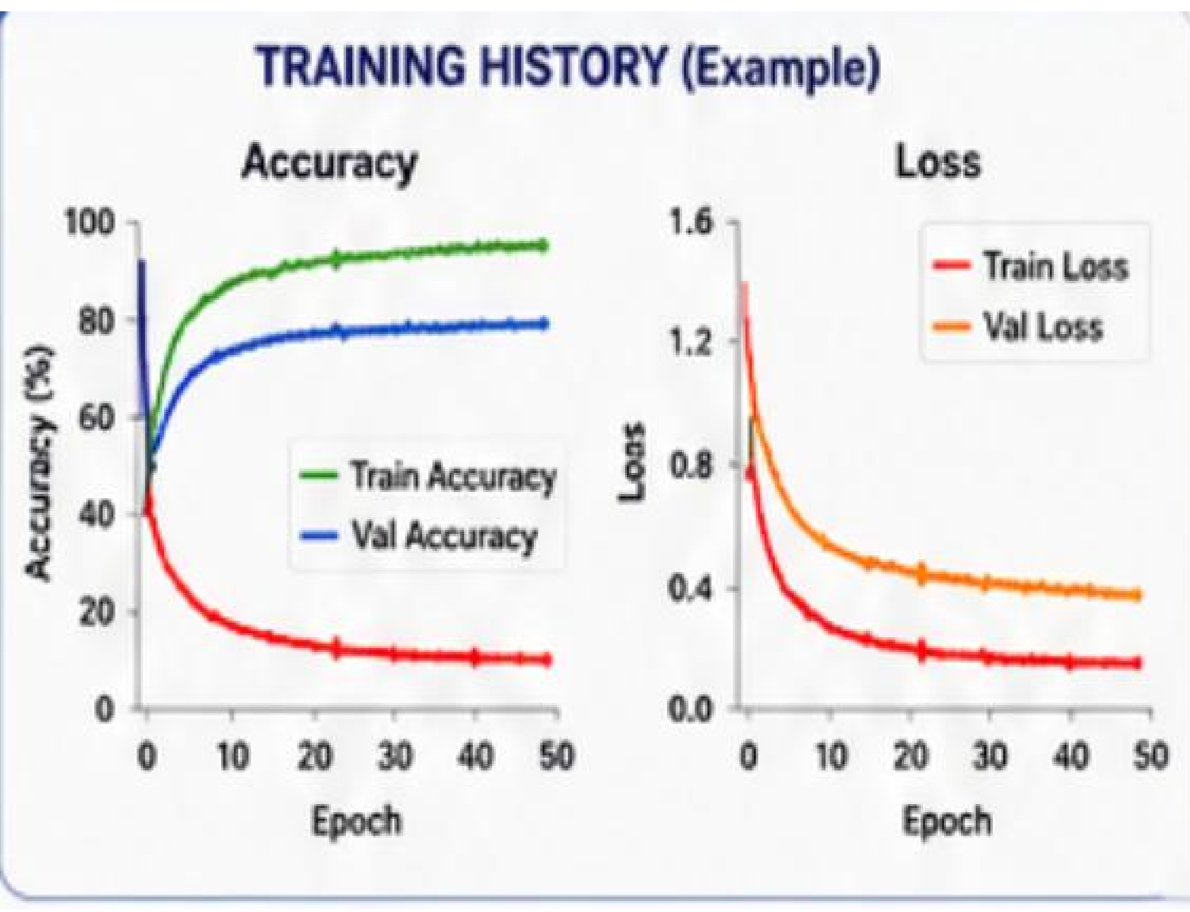


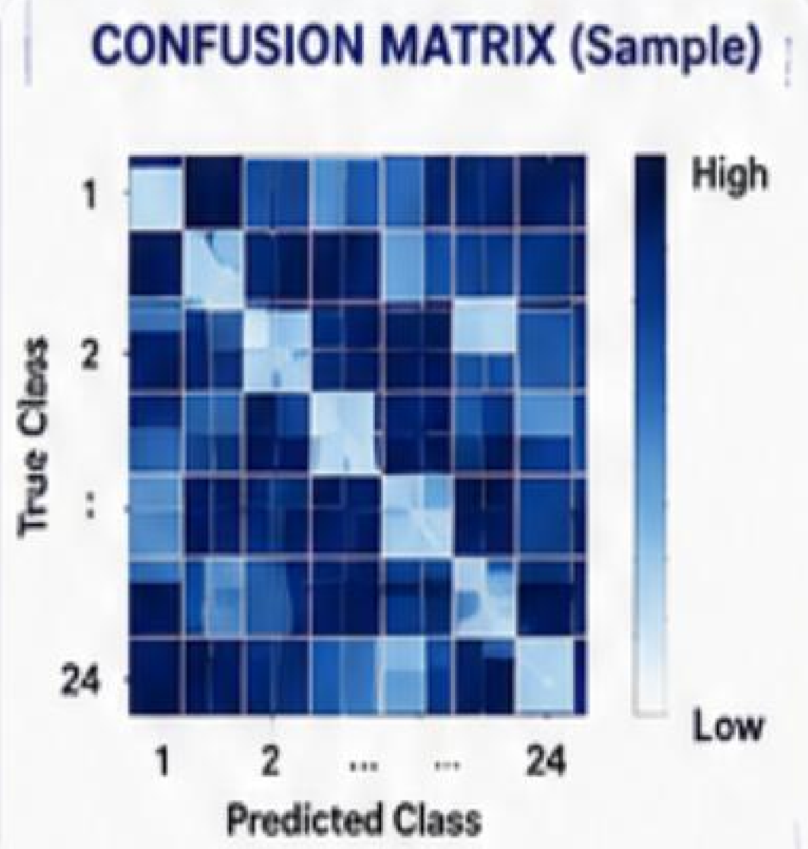
- Compound scaling: Uniformly balances depth, width, and resolution.
- Lightweight: High accuracy with minimal parameters.
- Mobile friendly: Uses MBConv blocks for fast processing.
- Edge optimized: Best choice for low-resource deployment.

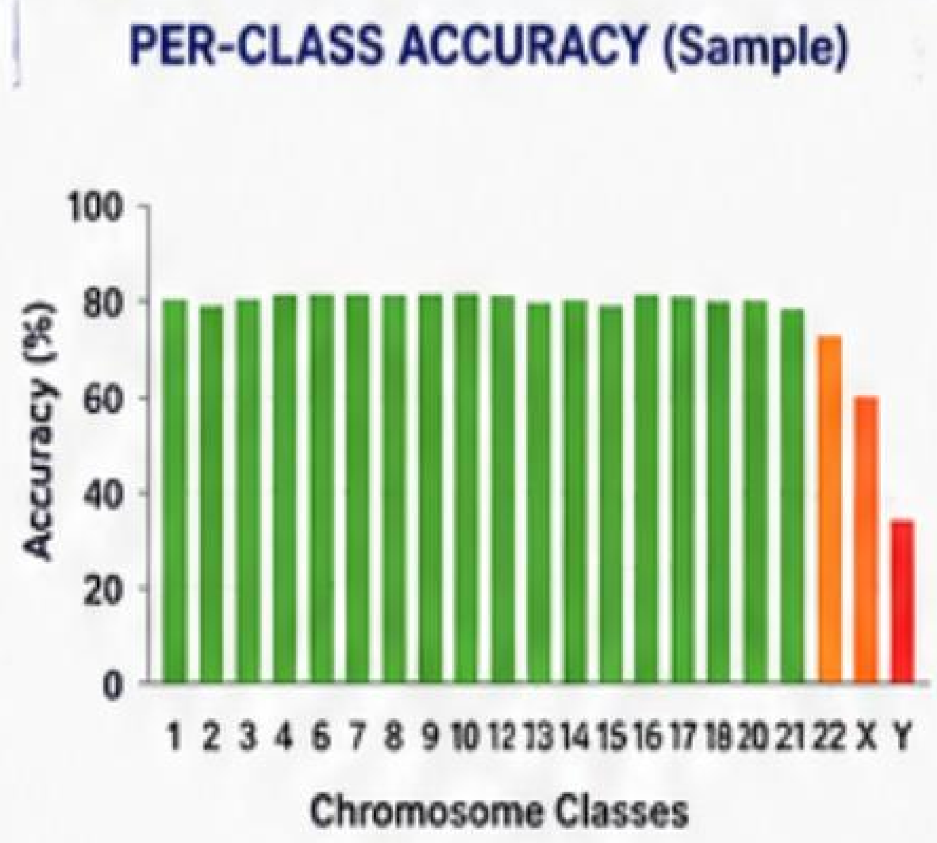
OUTPUT LAYER

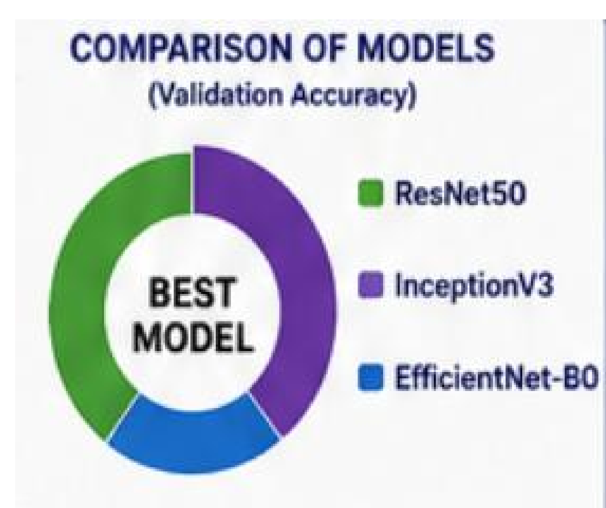


RESULTS & ANALYSIS

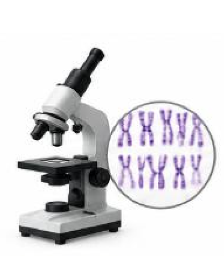






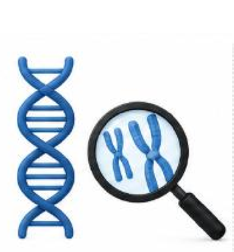


APPLICATIONS



CYTOGENETIC ANALYSIS
Automates identification of individual chromosomes to assist cytogenetic workflows.

KARYOTYPE ANALYSIS
Supports automated karyotyping by classifying chromosomes into standard 24 classes.



GENETIC RESEARCH
Facilitates large-scale genetic studies and chromosome analysis for research purposes.

GENETIC DISORDER DETECTION
Helps in detecting chromosomal abnormalities and supporting genetic disorder diagnosis.

BIOMEDICAL IMAGE ANALYSIS
Demonstrates the use of deep learning for high-accuracy biomedical image classification.

AUTOMATED SCREENING SUPPORT
Enables automated pre-screening of chromosome images, reducing manual effort and time.

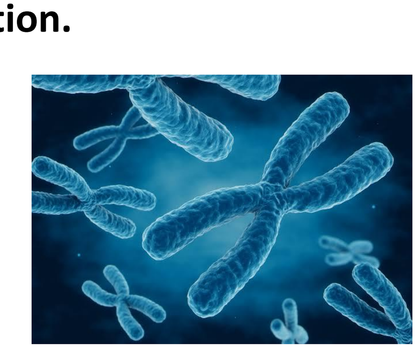
EDUCATION & TRAINING
Useful for teaching and training in cytogenetics and machine learning applications.

DATA MANAGEMENT & ARCHIVING
Helps in organizing, storing and retrieving large volumes of chromosome image data.

CONCLUSION

Deep learning can be effectively applied to classify individual chromosome images into 24 classes (1–22, X, Y). The proposed pipeline combines chromosome cropping, CLAHE, Gaussian filtering, sharpening, and data augmentation to improve image quality and model generalization. ResNet50, InceptionV3, and EfficientNet-B0 are evaluated to identify an effective architecture for accurate chromosome classification.

Chromosome Class	Denver group
1 to 3	A
4 to 5	B
6 to 12,X	C
13 to 15	D
16 to 18	E
19 to 20	F
21 to 22,Y	G



KEY HIGHLIGHTS

✓ 24-class chromosome classification (1–22, X, Y)

✓ Single chromosome image as input

✓ Sequential preprocessing: Crop → CLAHE → Gaussian → Sharpening

✓ Data augmentation to reduce overfitting

✓ Evaluation of ResNet50, InceptionV3 & EfficientNet-B0

✓ Confusion matrix and per-class performance analysis

✓ Automated and scalable chromosome classification